STATISTICS 101 · RECITATION HOMEWORK

Homework 2: Percentiles

Name: ______________________________ Date: ____________________

Instructions: About 20 minutes. Line numbers up before you find a median — order is everything today. Part C uses the growth chart; plot carefully. Total: **22 points**.

Part A — Line them up

1. Nine preemies weigh (in grams): 1,100, 760, 940, 1,600, 880, 990, 820, 1,250, 900. (a) Write them in order, smallest to biggest. (b) What is the median? (c) The mean is about 1,027. Why is the mean higher than the median here? (5 pts)

Part B — Place in line

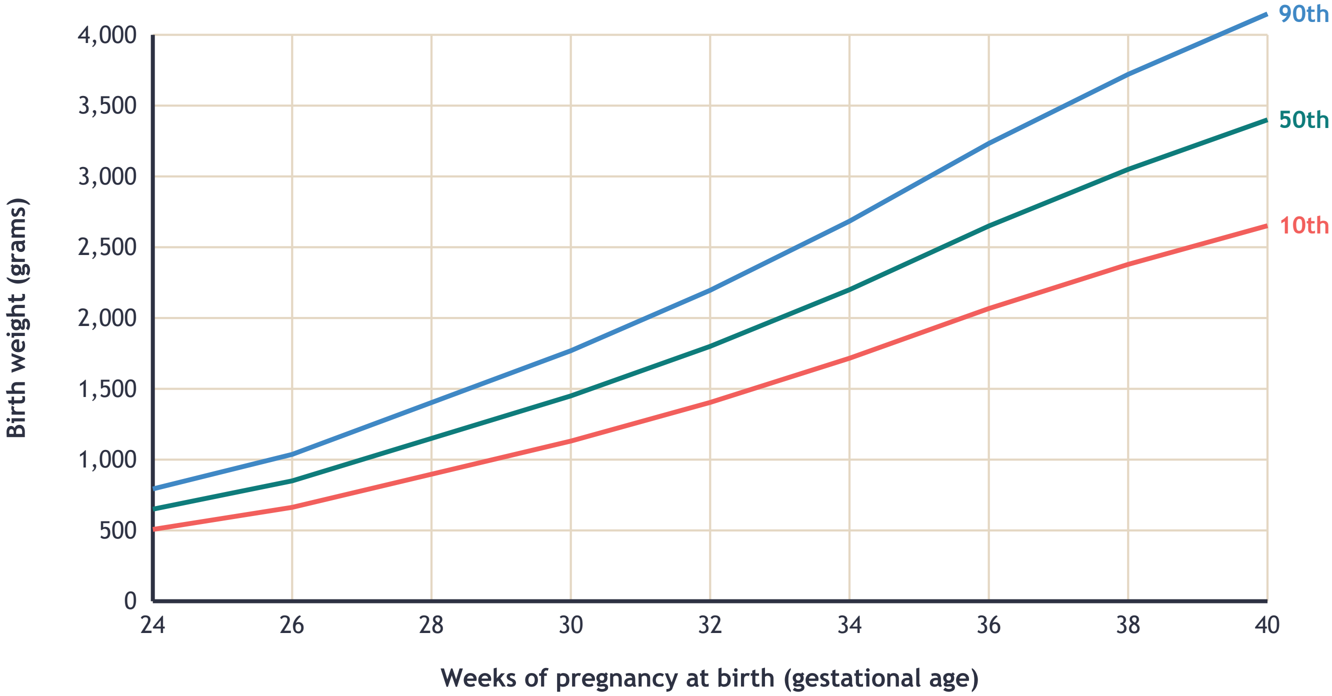
2. Twenty babies are lined up by weight. Turn each baby's position into a percentile. (Position ÷ 20 × 100.) (4 pts)

Baby	Position in line (from smallest)	Percentile
A	2nd	
B	10th	
C	15th	
D	19th	

3. Which of these is TRUE about percentiles? (1 pt)

- ☐ **A.** In any healthy group of babies, one in ten is at or below the 10th percentile.
- ☐ **B.** A baby below the 25th percentile is sick and needs treatment.
- ☐ **C.** The 90th percentile means the baby weighs 90 percent of the average.
- ☐ **D.** Percentiles go from 0 to 1,000, so the 50th is near the bottom.

Part C — On the chart



4. Plot these three babies on the chart as dots with their letters. Baby X: born at 28 weeks, 1,150 g. Baby Y: born at 32 weeks, 1,300 g. Baby Z: born at 36 weeks, 3,200 g. Then write which baby is near the 50th line, which is below the 10th, and which is above the 90th. (4 pts)

Part D — Following the curve

5. Baby M was born at 30 weeks weighing 1,450 g — right on the 50th line. Two weeks later, at 32 weeks, she weighs 1,450 g. (The 50th line at 32 weeks is about 1,800 g; the 10th is about 1,400 g.) What has happened to her place in line? Is this an alarm, even though 1,450 g is still 'a normal weight'? What would the team probably do? (3 pts)

Part E — Flashbacks

6. From NICU Tutor: a baby born weighing 1,300 g is called what kind of low birth weight? (1 pt)

7. From last week's Statistics: what is the mean of 110, 120, 130, and 160? (1 pt)

Part F — Think like a doctor

8. Two babies in the follow-up clinic are both at the 5th percentile for weight today. Baby P has been at the 5th percentile at every visit since birth; both parents are small. Baby Q was at the 50th percentile a month ago. Which baby worries you, and why? What does this tell you about what a percentile can and can't say? (3 pts)

HOME LAB (optional, not graded): Your percentile in the family

You need: a ruler or tape measure and everyone at home (plus friends or neighbors if you can catch them — more people makes a better line)

1. Measure every person's hand span: thumb tip to pinky tip, fingers spread wide. Write each one down with the person's name.
2. Line the numbers up from smallest to biggest. Find the median (the middle one, or halfway between the two middles).
3. Find YOUR position in line. Percentile = your position ÷ number of people × 100.
4. Now measure heights and do it again. Did your percentile change?
5. Bonus: add a grown-up's shoe or a stuffed animal's paw and watch what happens to the mean but not the median.

What I saw:

Done! Professor's initials: _________

Score: __________ / 22 points

Professor's comment: ___________________________________